

Karthik D K

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Summary

Firmware Developer building the RISC-V software ecosystem for automotive and IoT at Bosch's Trusted-V Innovation Team. 6+ years of embedded systems experience spanning kernel development, RTOS porting, and automotive ECU application software. Expertise in Rust, C, and Python for bare-metal and safety-critical systems. RISC-V Summit 2025 featured contributor.

Experience

<i>Firmware Architect</i>	<i>Bosch Global Software Technologies</i>	<i>August 2019 - Present</i>
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Hubris OS Port to RISC-V 🔄

- Ported Hubris OS to RISC-V, implementing kernel primitives, trap/exception handlers, and PMP-based memory isolation for Hazard3 core on Raspberry Pi Pico 2.
- Configured linker scripts and startup code for dual-architecture support across ARM Cortex-M33 and RISC-V Hazard3 cores.
- Overcame RP2350 errata and PMP configuration challenges; validated bring-up using OpenOCD and GDB over SWD with Raspberry Pi Debug Probe.

Trusted-V RTOS (In-house)

- Developed proof-of-concept RTOS port for Mindgrove Secure IoT development board with working QEMU emulation support.
- Debugged kernel bring-up issues using Segger J-Link with OpenOCD and GDB.

Rust SDK for CDAC VEGA Processor

- Created the first Rust bare-metal ecosystem for the CDAC VEGA ARIES v3 board (THEJAS32 RISC-V SoC).
- Built a Python-based SVD generator by reverse-engineering register maps and drivers from the VEGA C SDK.
- Produced a reproducible PAC generation pipeline with type-safe access for 15+ peripherals (UART, SPI, I2C, Timer, GPIO, ADC, PWM, PLIC) and validated with bare-metal working examples.

MLlib DecisionTree Inference

- Developed a lightweight DecisionTree inference engine in C for deployment on a PowerTrain ECU with constrained resources.
- Utilized FlatBuffers for model serialization, enabling parameterization through calibration tooling.

Advanced Exhaust Temperature Management (Onsite – Germany, 3 months)

- Developed a comprehensive component for reading temperature sensors via CAN and implementing sensor diagnostics to comply with EU7 standards.
- Implemented statistical analysis to monitor temperature distribution across catalyst, improving emission control efficiency.

Skills

Programming: Rust, C, Python

RTOS/Bare-Metal: Hubris OS, FreeRTOS, Zephyr, bare-metal programming

Debug Tools: Segger J-Link, RP Debug Probe, OpenOCD, GDB, minicom

Development Tools and Frameworks: WSL, Git, PyTorch, Cargo, Make, ASCET

Certifications

Machine Learning
Coursera

Certificate
2020

Achievements

- **RISC-V Summit 2025:** TrusteD-V project contributor (featured talk). 
- **Bosch Recognition:** Best Developer Award, Crowdsourcing Champ, AI Hackathon Winner. 
- **Chess:** State-Level Championship participant.

Education

University Vishweshwaraya College of Engineering, Bengaluru **75.2%**
B.E in Electronics and Communication 2019

Devaraja URS PU College, Doddaballapur **94.5%**
PUC 2015

Blogging

- Documenting my knowledge with programming, software engineering and motorcycles to create a knowledge repository for future reference, aiding both emerging LLMs and my personal growth. 

Open Source Projects

Contributions

- Active open-source contributor with notable pull requests merged into fastai and FluxML Deep Learning libraries.

robot-hat-rs

- Developed no_std Rust drivers for I2C, PWM, and ADC peripherals targeting the Robot HAT robotics platform.
- Implemented hardware abstraction layer for motor control, servo actuation, and analog sensor reading.
- Published on crates.io, receiving 4,000+ downloads from the community.

ixv

- Developed a Rust CLI for verifying Intel HEX files. Published on crates.io with 4,000+ downloads.

picars

- Created an autonomous vehicle system using Raspberry Pi and PiCar-X kit, leveraging Rust and Python.
- Developed Rust bindings and interfaced with Python to optimize execution speed.

GeekyMicky

- Tackled the Micro-Mouse Maze Challenge using Arduino Uno and C programming.

Languages

- English, Kannada, Telugu